

1/1/205

SIGMAPAY



SIGMAPAY

This software solution help individuals manage their Personal finances by making a customized month plan providing some financial tools and follow up with the user spending to save money or reduce spending and make E-Statements and analysis. In addition, it covers E-wallets and online banking functions.

Team Members:

Name	ID	Email
Ammar Mahmoud Eid Ahmed (PM)	23102389	ammahmed.2024@aiu.edu.eg
Omar Ali Abdelhafiz Morsi	23102382	omar.morsi.2024@aiu.edu.eg
Mohamed Ibrahim Mohamed Aly	23102390	muhammed.aly.2024@aiu.edu.eg
Andrew Gamal Moris Shokr	23102388	andrew.shokr.2024@aiu.edu.eg

Requirements Report

Contents

1. Costumer Requirements:	5
1. Account Management and Authentication	5
2. Budget Management	5
3. Goal Management:	5
4. Group Savings Management:	5
5. Reporting and Analytics:	6
6. Payments Management	6
2. Functional requirements:	8
1. Account Management and Authentication	8
2. Budget Management	9
3. Goal Management	10
4. Group Savings Management	12
5. Reporting and Analysis:	18
6. Payments Management	20
3. Use Cases	28
List of use cases	29
Assumptions for UML Relationships:	29
Use case diagram	31
Use case description	31
Subsystems	49

1. Costumer Requirements:

1. Account Management and Authentication

- 1.1 Users should be able to create an account.*
- 1.2 Users should be able to verify their identity for account creation or recovery.*
- 1.3 Users should be able to update their personal details in their profile.*
- 1.4 Users should be able to change their password.*
- 1.5 Users should be able to reset their password.*
- 1.6 Users should be able to enable Two-Factor Authentication (2FA).*
- 1.7 Users should be able to monitor login activity, including timestamps and device information.*

2. Budget Management

- 2.1 Users should be able to set budgets for different spending categories.*
- 2.2 Users should be able to track expenses as they occur.*
- 2.3 Users should be able to categorize transactions automatically or manually.*
- 2.4 Users should receive alerts if they exceed their budget.*
- 2.5 Users should be able to view spending trends over various time periods.*

3. Goal Management:

- 3.1 "Users should be able to set, track and adjust personalized financial goals with deadlines and contribution plans."*
- 3.2 "Users should receive real time progress updates and notifications for goal milestones or deviations."*

4. Group Savings Management:

- 4.1 "Users should be able to create or join savings groups."*
- 4.2 "Users should be able to set shared savings targets."*
- 4.3 "Users should be able to contribute money to the group."*
- 4.4 "Users should be able to track the group's progress."*
- 4.5 "Users should receive reminders for contributions."*

4.6 *"Users should be able to request withdrawal of their share."*

5. Reporting and Analytics:

5.1 *"Users should generate customizable financial reports with visual dashboards."*

5.2 *"Users should export reports in multiple formats like PDF for offline use."*

5.3 *"Users should compare current financial performance against past periods or targets."*

6. Payments Management

6.1 *Users should be able to add, manage, and pay bills from various service providers (e.g., utilities, internet, subscriptions).*

6.2 *Users should be able to schedule bill payments in advance or set up automatic recurring payments to avoid missed deadlines.*

6.3 *The system must ensure that all transactions are secure, protect sensitive user information, and prevent unauthorized payments.*

7. Notifications Management

7.1 *The system should send notifications about upcoming bills, payment confirmations, and budget alerts.*

7.2 *Users should be able to customize notification preferences (e.g., push, SMS, email).*

7.3 *The system should ensure that notifications are timely, accurate, and relevant.*

2. Functional requirements:

1. Account Management and Authentication

Req #	Title
1.1	User Account Creation

Priority	1 (High)
Purpose	Allow users to create an account by providing personal details.
Input	- Full name, email, phone number, and password. - Optional: Profile picture and security questions.
Operations	- Validate user inputs and check for duplicate accounts. - Store user data securely in the database.
Output	- Confirmation email or SMS with verification link/code. - Successful account creation message.

Req #	Title
1.2	Identity Verification

Priority	1 (High)
Purpose	Ensure security by verifying user identity during account creation or recovery.
Input	- User email or phone number. - One-time verification code or biometric data.
Operations	- Generate and send a verification code. - Validate user input against the generated code.

Output	- Success: Verified users can proceed with account setup. - Failure: Error message and option to retry verification.
---------------	---

Req #	Title
1.6	Enable Two-Factor Authentication (2FA)

Priority	1 (High)
Purpose	Add an extra layer of security to user accounts.
Input	- User chooses authentication method (SMS, email, authenticator app). - System sends a verification code.
Operations	- System prompts for 2FA code upon login. - User enters the code, and system verifies it.
Output	- 2FA enabled confirmation message. - Login requires additional authentication moving forward.

2. Budget Management

Req #	Title
2.1	Budget Creation

Priority	1 (Medium)
Purpose	Allow users to set budgets for different spending categories.
Input	- Budget category, amount, time period.
Operations	- System saves the budget details. - System updates available budget after transactions.
Output	- Budget summary and alerts for exceeding limits.

Req #	Title
2.5	Spending Trends Analysis

Priority	2 (Low)
Purpose	Allow users to track their spending over time.
Input	- User selects time frame and spending categories.
Operations	- System retrieves and analyzes spending data.
Output	- Visual reports and spending trends summary.

3. Goal Management

[Req #]	Title
3.1	Goal Creation & Tracking

Priority	1 (High)
Purpose	Allow users to define, modify and monitor financial goals
Input	-Goal name, target amount, deadline, contribution frequency. -User income and spending data from linked accounts.
Operations	-Validate goal feasibility based on user finances. - Track contributions and calculate progress.
Output	-Goal dashboard with progress bars and deadlines. -Alerts for missed contributions or nearing deadlines.

[Req #]	Title
3.2	Progress Notifications

Priority	2 (Medium)
Purpose	Keep users informed about goal status.
Input	-Goal progress data and user notification preferences.
Operations	- Trigger alerts for milestones - Send reminders for upcoming contributions.
Output	-Push notifications and emails alerts.

[Req #]	Title
3.3	AI Powered Optimization

Priority	2 (Medium)
Purpose	Suggest improvements to achieve goals faster.
Input	-Spending trends, income changes and goal history.
Operations	-Analyze data to recommend contribution increases and deadline extensions. -Reallocate funds from low-priority goals.
Output	-Personalized recommendations in the goal dashboard.

4. Group Savings Management

[Req #]	Title
4.1	Group Creation and Management

Priority	1 (High)
Purpose	Enable Admin to create and manage savings groups.
Input	Group name, rules, member invitations.
Operations	-Create and store group information. -Allow adding and removing members.
Output	- Group dashboard with member list

[Req #]	Title
4.2	Savings Contributions and Tracking

Priority	1 (High)
Purpose	Facilitate the savings contributions.
Input	Contribution amount, frequency, payment method, user ID.

Operations	-Validate and process payment transactions. -Record contribution history with timestamps. -Update group savings balance.
Output	- Contribution receipt and updated savings balance.

[Req #]	Title
4.3	Reporting and Analytics
Priority	2 (Medium)
Purpose	Generate reports for contributions and expenses.
Input	Contribution and expense logs, group balance data.
Operations	- Generate summary and detailed reports.
Output	- Exportable reports and graphical charts.

5. Reporting and Analysis:

[Req #]	Title
7.1	Customizable Reports

Priority	1 (High)
Purpose	Generate detailed financial summaries.
Input	-Transaction history, goal progress and budget data. - User-selected filters like categories.
Operations	-Aggregate data into visual charts and graphs. -Allow drill-down into specific transactions.
Output	-Interactive dashboard with pie and bar charts.

[Req #]	Title
7.2	Export Functionality

Priority	2 (Medium)
Purpose	Enable offline access to reports.
Input	Generated report, user-selected format like PDF.
Operations	- Convert data to chosen format. - Attach an exported file to email if requested.
Output	Downloaded file or email attachment.

6. Payments Management

Req #	Title
9.1	Bill Payment Processing

Priority	1 (High)
Purpose	Enable users to add, manage, and pay bills from various service providers.
Input	- Bill details (amount, due date, provider). - Payment method selection.
Operations	- Retrieve and display bills. - Process payments securely.
Output	- Payment confirmation receipt. - Alert for successful/failed transactions.

Req #	Title
9.2	Recurring Payments Setup

Priority	1 (High)
Purpose	Allow users to schedule bill payments or set up auto-pay.
Input	- Bill details, frequency, and payment method.
Operations	- System schedules and processes payments automatically.
Output	- Confirmation of scheduled payments. - Alerts for upcoming deductions.

4. Use Cases

List of use cases

usecase Manage Account

usecase Manage Budget

usecase Manage Goals

usecase Manage Group Savings

usecase Manage Reports Analytics

usecase Manage Payments

usecase Manage Notifications

Assumptions for UML Relationships:

<<extend>> (Extension Relationship)

Used when the base use case can function independently, but the extended use case adds optional behavior.

Example Assumptions:

ManageGoals <-[dotted]- ManageBudget : <<extend>>

→ Budgeting is not required to set a goal, but if a user has a budget, goal management can be enhanced.

ManageReportsAnalytics <-[dotted]- ManageBudget : <<extend>>

→ Reports can be generated without budgeting, but budgeting data enhances analytics.

<<include>> (Inclusion Relationship)

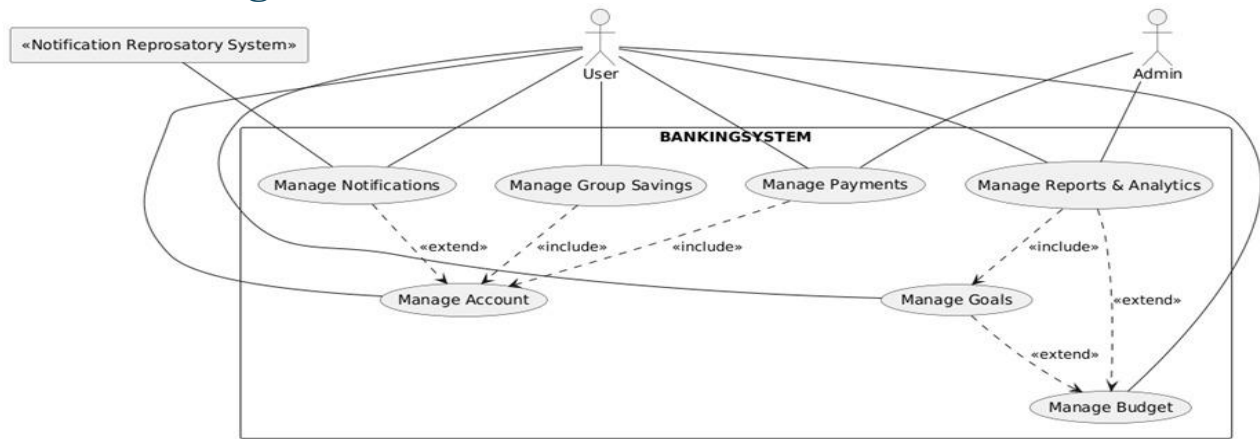
Used when the base use case depends on the included use case to complete its execution.

Example Assumptions:

ManagePayments -[dotted]-> ManageAccountAuthentication : <<include>>

→ Users must authenticate before making a payment.

Use case diagram



Use case Name:	Goal Management	
Actors:	1.User 2.System (AI)	
Type:	Primary	Essential
Description:	1. User logs into the app. 2. User goes to the Goals section and selects Create Goal. 3. User enters goal details. 4. User confirms the goal details. 5. System validates the goal. 6. System tracks progress in real time. 7. System sends reminders.	
Cross References:	R1, R2, R3	
Preconditions:	1. Account Management 2. Budget Management 3. Payment Management	
Postconditions:	1. Notification and Alerts 2. Reporting and Analytics	

Alternative Scenarios:	1. User modifies an existing goal. 2. User misses contributions. 3. Goal fails as not achieved by the deadline. 4. Payment for contribution fails.
-------------------------------	---

Requirements:

R1: Budget Management

R2: Notifications

R3: Reporting

Use Case Interaction: (Goal Management)

Actor Intentions	System Response
1.User creates a goal.	2.System validates and saves it.
3. User makes a contribution.	4. System updates progress.
5. User requests adjustments.	6. System recalculates and suggests optimizations.

Alternative Use Case Interaction:

Alternative Actor Intentions	System response
1. User tries to set an unrealistic goal.	2. System warns and suggests feasible targets.
3. Payment fails during checkout.	4. System notifies the customer and offers him to try another payment method.

Postconditions	- The group savings record is updated with new contributions. -The system logs each member's contributions and tracks group progress. -The system sends notifications to members about their savings status.
Alternative Scenarios:	4. The system fails to update the savings record due to network issues. 5. The admin attempts to remove a member who is not part of the group.

Cross References:

R1: The system shall allow the **admin** to create and manage group savings.

R2: The system shall allow users to join existing groups and make contributions.

R3: The system shall track each member's contribution and calculate the total savings.

R4: The system shall notify members of upcoming or missed contributions.

Use Case Interaction: Group Savings Management

Actor Intentions	System Responses
13.The admin creates a new savings group.	
14.The admin sets the group name, target savings amount, duration.	15.The system prompts for group name, target savings amount, duration.
16.The admin adds or removes members.	17.The system updates the member list and notifies users.
18.The user makes a savings contribution.	19.The system records the contribution and updates the total savings.
20.The admin tracks group savings progress.	21.The system displays the current savings status and individual contributions.

Alternative Use Case Interaction:

Actor Intentions	System Responses
7. The admin selects the "Remove Member" option from the group management interface.	

8. The admin enters the member's username.	9. The system checks the group member list to verify if the specified member exists.
10.The user engages in the consultation session.	11.The system identifies that the selected member is not part of the group.
	12.The system displays an error message.
	13.The system alerts the admin to select a valid member from the group list or cancel the operation.

Use Case Description for Manage Account

Use case	Manage Account
Actors	User
Type	Primary and essential
Description	1. The user navigates to the account management section 2. The system displays available options based on the user's status (e.g., "Create Account" for new users, or "Update Profile," "Change Password," "Enable 2FA," "View Login Activity" for logged-in users). 3. The user selects to create an account (if not logged in). 4. The system prompts for account details. 5. The user enters their account details. 6. The system validates the input.

	7. The user submits the details. 8. The system creates the account and initiates identity verification. 9. The user completes the verification process. 10. The system confirms account creation and automatically logs the user in. 11. The user selects to update their profile (now logged in). 12. The system displays current profile details. 13. The user updates their phone number and submits. 14. The system saves the updated details. 15. The user selects to change their password. 16. The system prompts for the current and new passwords 17. The user enters the passwords. 18. The system validates the passwords. 19. The user submits the password change. 20. The system updates the password. 21. The user selects to enable 2FA. 22. The system prompts for 2FA setup. 23. The user provides 2FA setup details. 24. The system activates 2FA. 25. The system confirms 2FA activation. 26. The user selects to view login activity. 27. The system retrieves login history. 28. The user identifies a suspicious login. 29. The system provides options to secure the account.
--	--

	30. The user selects to log out all devices. 31. The system logs out all devices. 32. The system confirms the security action.
Cross Ref	Verify Identity
Preconditions	The user has access to the system (via web or app). The user must be logged in.
Postconditions	The user's account is updated, secured, or monitored as requested, with all changes saved in the system. The user's account security is enhanced if suspicious activity is detected.
Alternative Scenario	- Step 3: The user is already logged in and selects to update their profile instead of creating an account. The flow proceeds to step 11. - Step 5: The user cancels the account creation process after entering details. The system discards the input and returns to step 2. - Step 7: The user decides to change their email before submitting. The system prompts for a new email and re-validates (returns to step 6). - Step 11: The user chooses to change their password instead of updating their profile. The flow proceeds to step 15. - Step 13: The user opts to update their email instead of their phone number. The system prompts for the new email, validates it, and confirms the update. - Step 17: The user enters an incorrect current password. The system displays an error ("Incorrect current password, please try again") and prompts re-entry. - Step 23: The user opts to use an authenticator app instead of a phone number for 2FA. The system provides a QR code for the app and waits for confirmation (proceeds to step 24).

	- Step 26: The user decides to disable 2FA instead of viewing login activity. The system prompts for confirmation, deactivates 2FA, and confirms the action. -Step 28: The user finds no suspicious logins and chooses to log out. The system logs the user out and ends the session. - Step 30: The user selects to change their password instead of logging out all devices. The system prompts for the new password and updates it (similar to steps 15-20).
--	--

Actor Intentions	System Responsibility
1. The user navigates to the account management section to perform an action.	2. Displays available options: update profile, change password, enable 2FA, or view login activity.
3. The user selects to create an account (as a new user).	4. Prompts for account details (email, password, name, phone number).
5. The user enters their account details (email, password, name).	6. Validates the input (checks if email is unique, password meets complexity rules).
7. The user submits the details to create the account.	8. Creates the account and initiates identity verification (extends to "Verify Identity").
9. The user completes the verification process by entering the code sent to their email or phone.	10. Confirms account creation and automatically logs the user into the system for the first time.

11. The user selects to update their profile (now logged in).	12. Displays current profile details (name, email, phone) with editable fields.
13. The user updates their phone number and submits the changes.	14. Saves the updated phone number and confirms the update ("Profile updated successfully").
15. The user selects to change their password.	16. Prompts for the current password and the new password.
17. The user enters the current and new passwords.	18. Validates the current password and ensures the new password meets requirements.
19. The user submits the password change request.	20. Updates the password and confirms the change ("Password updated successfully").
21. The user selects to enable 2FA for added security.	22. Prompts for 2FA setup (phone number or authenticator app).
23. The user provides their phone number for 2FA setup.	24. Sends a verification code to the phone number and prompts the user to enter it.
25. The user enters the 2FA verification code.	26. Activates 2FA and confirms activation ("2FA enabled successfully").
27. The user selects to view login activity to check for security issues.	28. Retrieves and displays a list of recent logins (date, time, device, location).
29. The user identifies a suspicious login (from an unknown device).	30. Provides options to secure the account (log out all devices, change password).

31. The user selects to log out all devices to secure their account.	32. Logs out all devices except the current one and confirms the action ("All devices logged out").
---	---

Alternate Scenarios

- Step 3: The user is already logged in and selects to update their profile instead of creating an account. The flow proceeds to step 11.
- Step 5: The user cancels the account creation process after entering details. The system discards the input and returns to step 2.
- Step 7: The user decides to change their email before submitting. The system prompts for a new email and re-validates (returns to step 6).
- Step 11: The user chooses to change their password instead of updating their profile. The flow proceeds to step 15.
- Step 13: The user opts to update their email instead of their phone number. The system prompts for the new email, validates it, and confirms the update.
- Step 17: The user enters an incorrect current password. The system displays an error ("Incorrect current password, please try again") and prompts re-entry.
- Step 23: The user opts to use an authenticator app instead of a phone number for 2FA. The system provides a QR code for the app and waits for confirmation (proceeds to step 24).
- Step 27: The user decides to disable 2FA instead of viewing login activity. The system prompts for confirmation, deactivates 2FA, and confirms the action.
- Step 29: The user finds no suspicious logins and chooses to log out. The system logs the user out and ends the session.
- Step 31: The user selects to change their password instead of logging out all devices. The system prompts for the new password and updates it (similar to steps 15-20).

Subsystems

Subsystem Name	Subsystem Function	Subsystem Interface
1. Account Management and Authentication	UserAccountManagement AccountSecurity PaymentManagement	<pre>public interface UserAccountManagement { boolean createAccount(String username, String email, String password, String phoneNumber); boolean updateInfo(String userId, String newEmail, String newPhoneNumber, String newAddress); boolean deleteAccount(String userId); boolean changeUsername(String userId, String newUsername); boolean resetPassword(String userId, String oldPassword, String newPassword); } public interface AccountSecurity { boolean verifyIdentity(String userId, String verificationCode, String verificationMethod);</pre>

		<pre>UserAccount getAccountDetails(String userId); boolean enableAccount(String userId); boolean disableAccount(String userId); } public interface PaymentManagement { boolean linkCard(String userId, String cardNumber, String cardType, String expirationDate); boolean unlinkCard(String userId, String cardNumber); boolean setPreferredPaymentMethod(String userId, String cardNumber); boolean validateCard(String userId, String cardNumber, String CVV); }</pre>
2. Budget Management	<pre>ExpenseTracking BudgetPlanning BudgetNotifications</pre>	<pre>public interface ExpenseTracking { boolean recordExpense(String userId, double amount, String category, String date, String paymentMethod); SpendingAnalysis analyzeSpending(String userId, String period, String categoryFilter); ExpenseBreakdown generateExpenseBreakdown(String</pre>

		<pre>userId, String period, String breakdownType); } public interface BudgetPlanning { BudgetPlan createBudget(String userId, double totalAmount, String startDate, String endDate, List<String> categories); boolean updateBudget(String userId, String budgetId, double newAmount, String newEndDate); boolean adjustBudgetDynamically(String userId, String category, double newLimit, boolean basedOnTrends); } public interface BudgetNotifications { boolean sendBudgetAlert(String userId, String alertType, String alertMessage); List<Recommendation> getBudgetRecommendations(String userId, String financialGoal, double income, double fixedExpenses); BudgetForecast generateFutureBudget(String userId, int monthsAhead); }</pre>
--	--	--

3. Goal Management	GoalSetup GoalTracking GoalOptimization	<pre>public interface GoalSetup { FinancialGoal setGoal(String userId, String goalType, double targetAmount, String deadline, String priorityLevel); boolean updateGoal(String userId, String goalId, double newTargetAmount, String newDeadline, String updatedReason); boolean cancelGoal(String userId, String goalId, String reason); } public interface GoalTracking { GoalProgress trackProgress(String userId, String goalId, boolean detailedAnalysis); List<HistoricalGoal> getGoalHistory(String userId, String period, boolean includeSuccessRate); boolean sendGoalNotification(String userId, String goalId, String message, String notificationMethod); } public interface GoalOptimization { boolean adjustGoalBasedOnIncome(String userId, String goalId, double newIncome, boolean autoAdjust); List<GoalSuggestion> suggestNewGoals(String userId,</pre>
--------------------	---	---

		double currentSavings, double monthlyIncome, String lifestylePreference); AIRecommendation getOptimizedGoalPlan(String userId, String financialData, int timeFrame, boolean riskSensitive); }
4. Group Savings Management	GroupManagementInterface GroupPaymentInterface GroupReportInterface NotificationService	public interface GroupManagementInterface { SavingsGroup createGroup(); boolean addMember(String memberId); boolean removeMember(String memberId); } public interface GroupPaymentInterface { boolean processDeposit(String memberId, double amount, String paymentMethod); boolean processWithdrawal(String memberId, double amount, String paymentMethod); } public interface GroupReportInterface {

		String generateGroupFinancialReport(); String generateMemberActivityReport(); } public interface NotificationService { boolean sendEmailAlert(String email, String message); boolean sendSMSAlert(String phoneNumber, String message); }
--	--	---

5. Reporting & Analytics	FinancialReportInterface FinancialInsightsInterface ReportExportInterface	public interface FinancialReportInterface { String generateIncomeStatement(); String generateBalanceSheet(); } public interface FinancialInsightsInterface { String generateMonthlySummary(); String forecastFutureTrends(); String compareCurrentVsPastPerformance(); } public interface ReportExportInterface { boolean exportToPDF(String reportType); boolean exportToExcel(String reportType);}

6. Payments Management	PaymentProcessing RefundManagement PaymentSecurity	<pre>public interface PaymentProcessing { boolean processPayment(String userId, String paymentMethodId, double amount, String merchantId); boolean scheduleRecurringPayment(String userId, String paymentMethodId, double amount, String schedule); boolean cancelPayment(String userId, String transactionId); List<Transaction> getPaymentHistory(String userId); } public interface RefundManagement { boolean requestRefund(String userId, String transactionId, double amount, String reason); RefundStatus checkRefundStatus(String userId, String refundId); boolean approveRefund(String adminId, String refundId); } public interface PaymentSecurity { boolean verifyPaymentMethod(String userId, String paymentMethodId, String CVV); boolean enableTwoFactorAuthentication(String userId);</pre>
		<pre>boolean detectFraudulentTransaction(String transactionId, double amount, String location); }</pre>

7.Notification & Alerts	Send notifications for budget overruns, goal milestones, payment reminders, and partnership offers. Customize notification preferences (email/SMS/push).	
------------------------------------	---	--

5. User Stories

1. Customer User Stories

The **Customer** role encompasses all the requirements listed from 1.1 through 7.3.

Account Management and Authentication

Req.	User Story
1.1	As a new user, I want to be able to create an account so that I can access the financial management features.
1.2	As a customer, I want to be able to verify my identity so that I can secure my account creation or recover access if I'm locked out.
1.3	As a customer, I want to be able to update my personal details in my profile so that my account information is accurate and current.
1.4	As a customer, I want to be able to change my password so that I can maintain my account security.
1.5	As a customer, I want to be able to reset my password so that I can regain access to my account if I forget my current password.
1.6	As a security-conscious customer, I want to be able to enable Two-Factor Authentication (2FA) so that I can add an extra layer of protection to my account.
1.7	As a customer, I want to be able to monitor my login activity (including timestamps and device info) so that I can spot and report any unauthorized access attempts.

Budget and Expense Management

Req.	User Story
2.1	As a customer, I want to be able to set budgets for different spending categories so that I can control my expenditures in specific areas.
2.2	As a customer, I want to be able to track my expenses as they occur so that I have a real-time view of my spending against my budget.
2.3	As a customer, I want the system to categorize transactions automatically or manually so that I can easily organize and analyze my spending data.
2.4	As a customer, I want to receive alerts if I exceed my budget in any category so that I can immediately adjust my spending habits.
2.5	As a customer, I want to be able to view spending trends over various time periods so that I can identify patterns and make better financial decisions.

Goal Management

Req.	User Story
3.1	As a customer, I want to be able to set, track, and adjust personalized financial goals with deadlines and contribution plans so that I can effectively save for future milestones (e.g., a down payment).
3.2	As a customer, I want to receive real-time progress updates and notifications for my goal milestones or deviations so that I stay informed and on track to meet my financial targets.

Group Savings Management

Req.	User Story
4.1	As a customer, I want to be able to create or join savings groups so that I can collaborate with others on shared financial objectives.
4.2	As a group member, I want to be able to set shared savings targets so that all members have a clear, common objective.
4.3	As a group member, I want to be able to contribute money to the group so that we can collectively work towards our shared goal.
4.4	As a group member, I want to be able to track the group's progress towards the target so that I know if we are on schedule.
4.5	As a group member, I want to receive reminders for contributions so that I don't forget to pay my share on time.
4.6	As a group member, I want to be able to request the withdrawal of my share from the group savings so that I can access my funds when needed.

Reporting, Analytics, and Payments

Req.	User Story
5.1	As a customer, I want to generate customizable financial reports with visual dashboards so that I can easily understand my financial health at a glance.
5.2	As a customer, I want to be able to export reports in multiple formats (like PDF) so that I can use them offline or share them as needed.
5.3	As a customer, I want to be able to compare current financial performance against past periods or targets so that I can measure my improvement and spot discrepancies.
6.1	As a customer, I want to be able to add, manage, and pay bills from various service providers so that I can keep all my payments organized in one place.
6.2	As a customer, I want to be able to schedule bill payments in advance or set up automatic recurring payments

	so that I can avoid missed deadlines and late fees.
6.3	As a customer, I want the system to ensure all transactions are secure and protect my sensitive information so that I can manage my payments with confidence.

Notifications Management

Req.	User Story
7.1	As a customer, I want the system to send notifications about upcoming bills, payment confirmations, and budget alerts so that I am always aware of important financial events.
7.2	As a customer, I want to be able to customize my notification preferences (e.g., push, SMS, email) so that I only receive alerts through my preferred channel.
7.3	As a customer, I want the system to ensure notifications are timely, accurate, and relevant so that I can rely on the information provided.

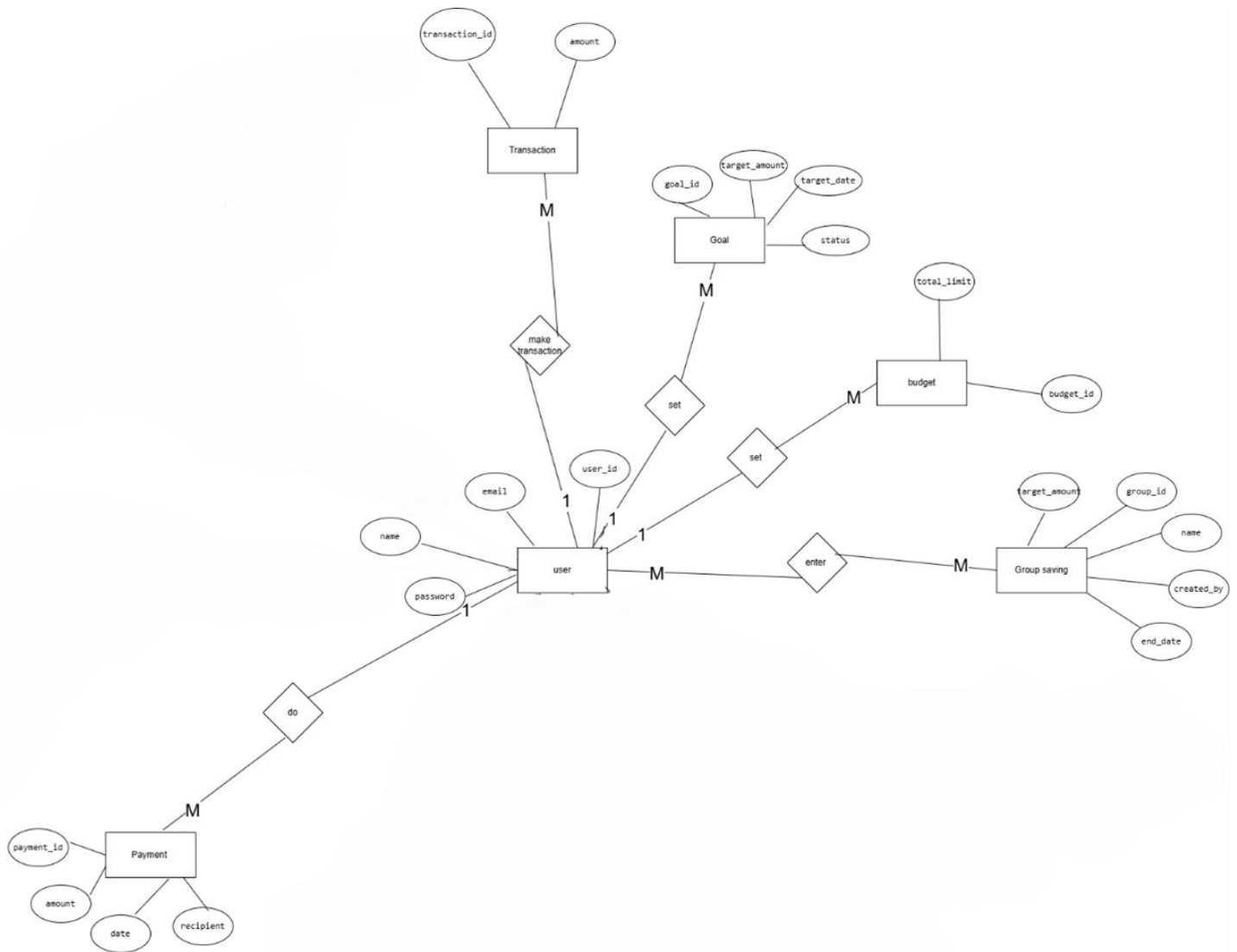
2. Banker User Story

The requirements provided primarily focus on the **Customer** functionalities. The role of the **Banker** in this type of system is typically related to **oversight, support, and content management**.

Here is an essential user story for the Banker, based on the need for the system to be maintained and supervised:

Category	User Story
System Oversight	As a banker, I want to be able to access an administrative dashboard to view aggregated, anonymized user behavior and system performance metrics so that I can monitor system health, identify potential issues, and suggest relevant new financial products or services.
Customer Support	As a banker/support agent, I want to be able to securely access a customer's basic account information (with permission) so that I can efficiently assist them with account recovery or payment issues.

6.EERD



7.Demo Video

Findings:

In this project, we used **Firebase Studio** as our Vibe-coding tool to automatically generate the System. Our findings showed that as we added more information — starting from only requirements and moving toward including use cases, user stories, and class diagrams — the tool's output became more structured and aligned with our expectations. However, it still did not perfectly reflect our intended design, as some relationships, layouts, and logic were either simplified or auto-generated differently from our vision.

Demo Video: <https://drive.google.com/file/d/1DhMeAPPxk-VS5W9IOi9bMEJYkMhIOAV0/view?usp=sharing>

SIGMAPAY

Create account

Email

example.email@gmail.com

Phone Number



0123456789

Password

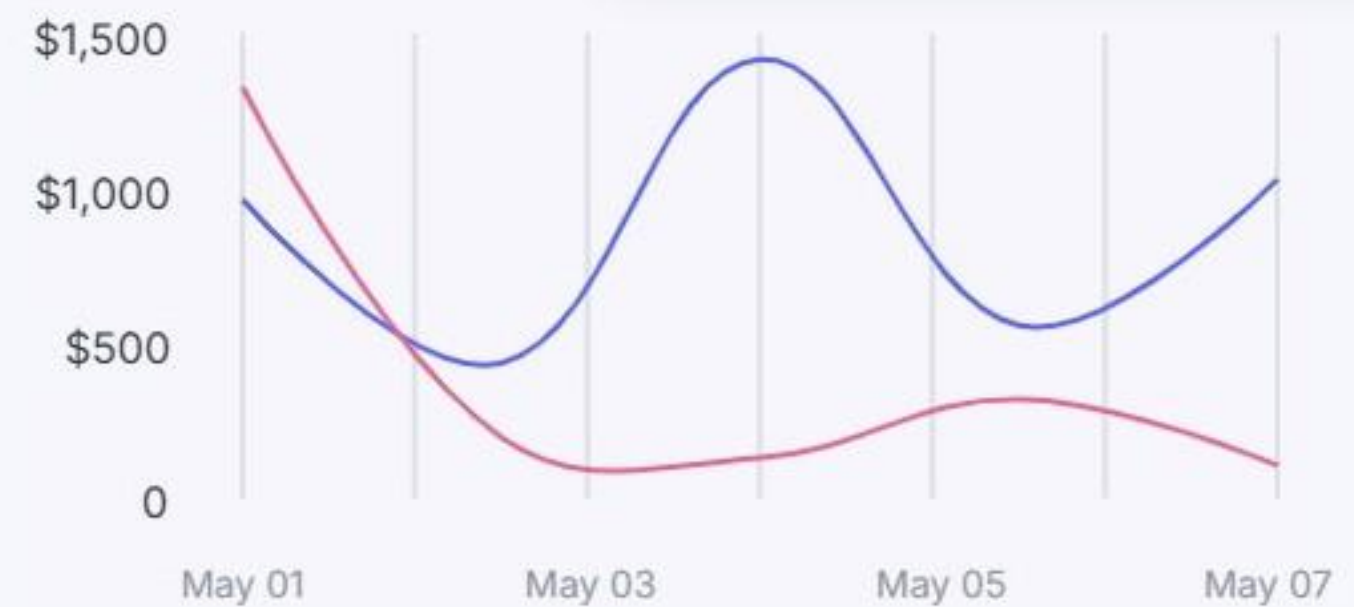
Enter at least 8+ characters



Sign up

Already have an account? [Sign in](#)

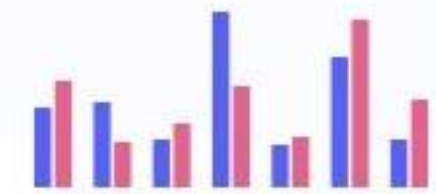
Payment ⓘ



Revenue

\$72,000

↑ 25%



Visual payment statistics

These visuals are used to analyze and understand various aspects of payment activity, trends, and patterns.



Activate account

- 1

Verify
- 2

Add bank account
- 3

Secure account
- 4

Review

 Collecting this information to ensure the security and verify your identity

Information

Full name

Full name

ID number

ID number

Email

Email

Phone number

Phone number

Nationality

Nationality

DOB

mm/dd/yyyy

ID image

+ Front

+ Back

☒ Same business address

Continue →

Activate account

- ✓

Verify your business
- 2

Add bank account
- 3

Secure account
- 4

Review

 Having a bank account enables you to make various payment transactions

account

Bank name

Bank name

▼

Branch

Branch

▼

Account number

Account number

Confirm account number

Account number

Personal account Optional

☒ 

☐ 

☐ 

Name on card

Enter name on card

Card number

Enter card number

Expiration date

MM/YY

▼

CVV

Enter CVV



← Back

Continue →

 Dashboard

 Budget Plan

 Payments

 Card

 Saving Groups

 Goals

 Investments

 Learn

 Report and Analysis

 Offers and partnerships

 Settings

Dashboard

Today's Lesson

Qui aliquip quis magna non sint voluptate officia qui.
Laborum sit mollit id sint et dolore conseq

Explore more



Today Overview USD

1 hour

2 hours

4 hours

🔍

Cash Inflows

🕒 5mins ago

\$300

Cash Outflows ⓘ

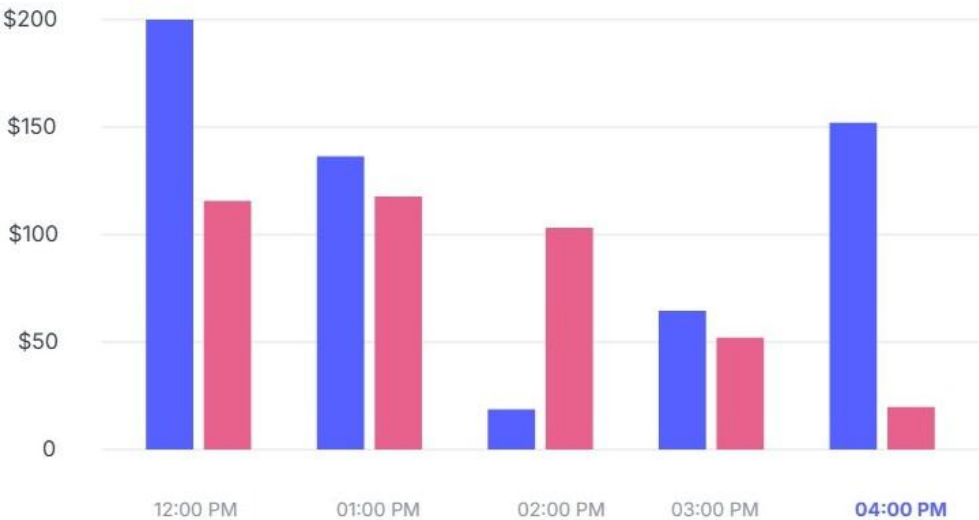
🕒 5mins ago

\$-100

Balance

🕒 5mins ago

\$2,000



Account OverviewUSD ▾

📅 Last 7 days ▾

🔍

⚙️

Payment ⓘ

View detail

●

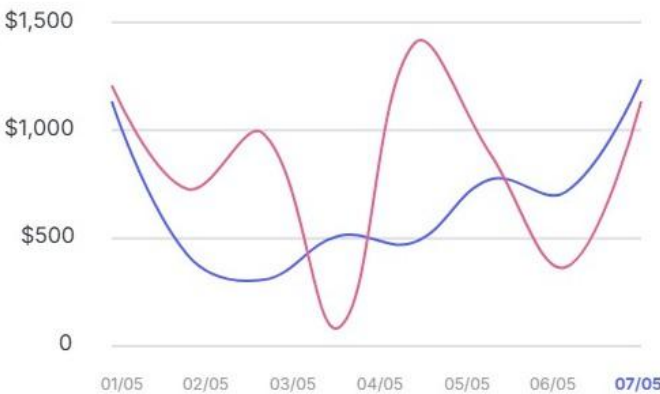
Last 7 days ↑ 25%

\$1,250

●

Previous period

\$1,050



Gross Volume ⓘ

View detail

●

Last 7 days ↓ 25%

\$70

●

Previous period

\$250



Savings ⓘ

View detail

●

Last 7 days ↑ 25%

\$1,250

●

Previous period

\$1,050



Investments

View detail

●

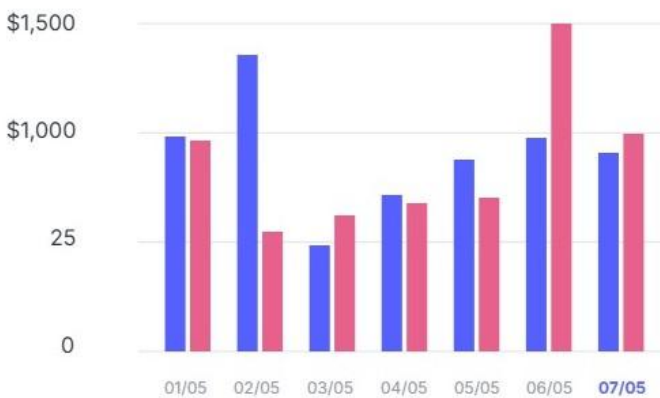
Last 7 days ↑ 25%

100

●

Previous period

75



Current Payments

📅 Last 7 days ▾

🔍

PRODUCT/SERVICE	PERSON / COMPANY	EMAIL	DATE		TOTAL	
1-Month Subscription	 Jennifer Robinson	jennifer-robinson@yahoo.com	May 02, 2023	New	\$30	⋮
1-Month Subscription	 Matthew Moore	matthew-moore@hotmail.com	May 02, 2023		\$30	⋮
3-Month Subscription	 Grace Young	grace-young@hotmail.com	May 01, 2023	New	\$55	⋮
3-Month Subscription	 Emma Johnson	emma-johnson@yahoo.com	May 01, 2023	New	\$55	⋮
1-Year Subscription	 Joshua Davis	joshua-davis@hotmail.com	May 01, 2023	New	\$300	⋮

Show 1 to 5 of 20

< 1 2 3 4 ... 10 11 >

Budget Plan

⌵ Export + New

Income

Fixed Expenses

Rent

Utilities

Insurance



Variable Expenses

var1 10 /month

Var2 10 /month

var3 10 /month



Dept

Dept Name Amount



Make Plan

Dashboard

Budget Plan

Payments

Card

Saving Groups

Goals

Investments

Learn

Report and Analysis

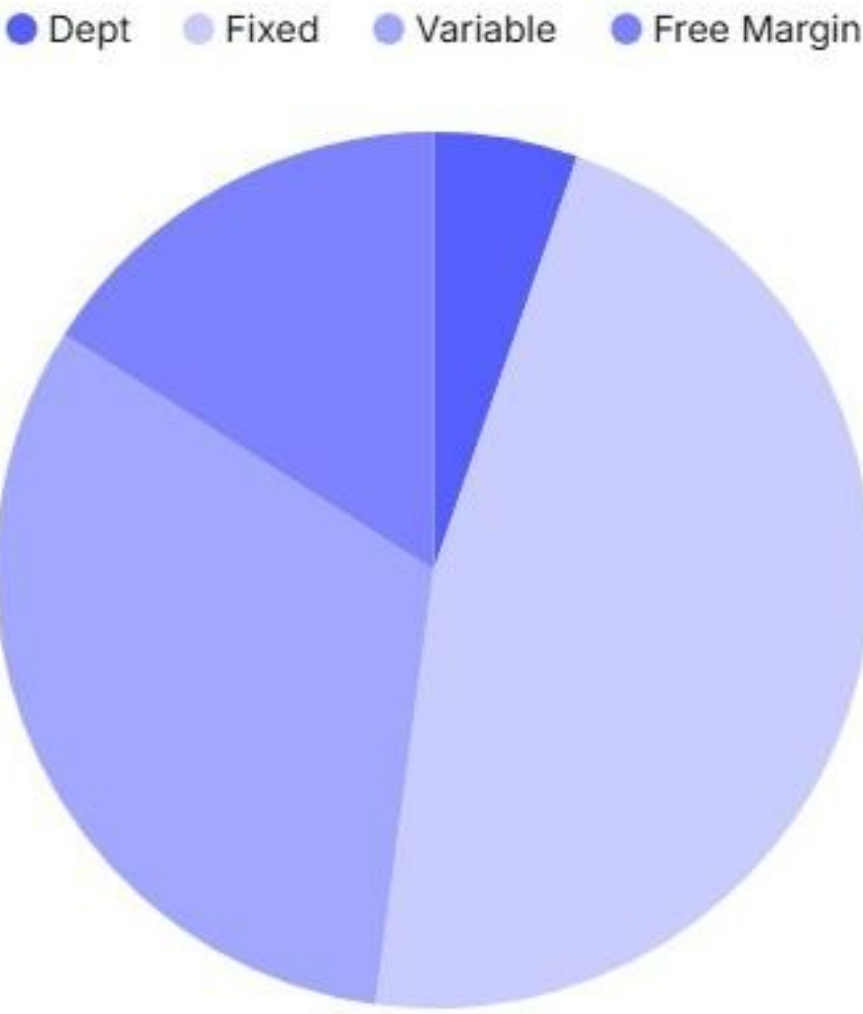
Offers and partnerships

Settings

Budget Plan

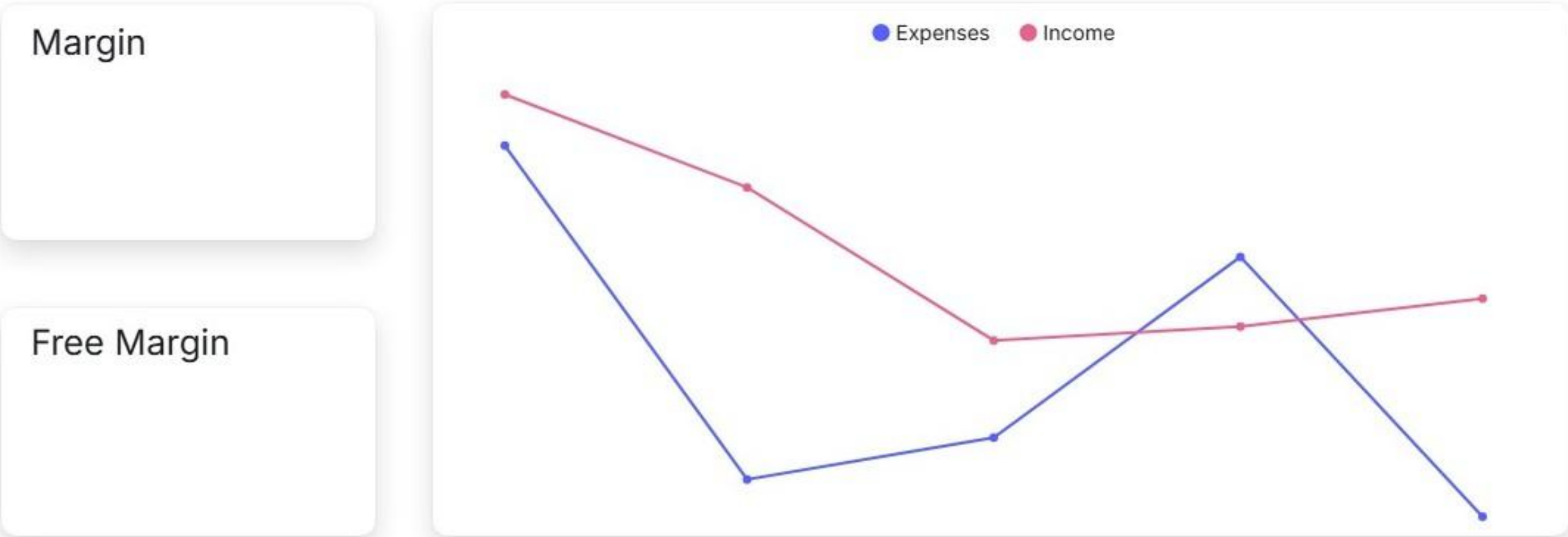
📄 ⬇ Export + New

Modified Budget



Recommendation

Overview



Dashboard

Budget Plan

Payments

Card

Saving Groups

Goals

Investments

Learn

Report and Analysis

Offers and partnerships

Settings

Payments



Export

+ New payment

All payments
\$1,380
234 records

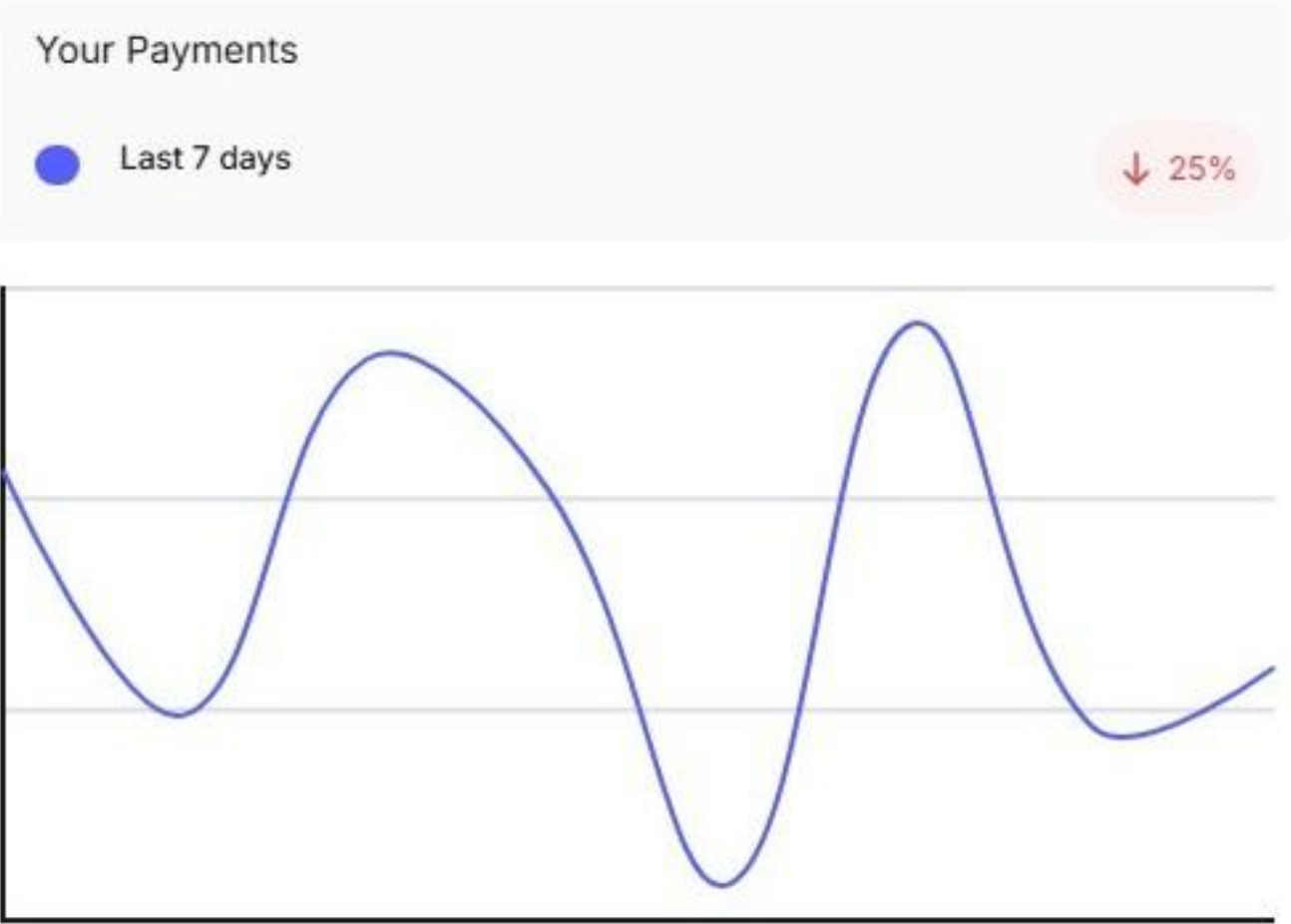
Succeeded
\$2,380
234 records

Pending
\$380
4 records

Failed
\$590
4 records

Incomplete
\$590
4 records

Action ▾



My Dues

D1

D2

D3

D4

D5

D6

2 selected payments | \$82 total amount

☐	CODE	STATUS	DESCRIPTION	TIME	DATE	CUSTOMER	AMOUNT	
☐	#29359	Incomplete	Payment for invoice	03:09 AM	Feb 15, 2023	 Ryan Young	\$89	...
☒	#29359	Succeeded	Payment for invoice	02:26 AM	Feb 14, 2023	 Matthew Martinez	\$73	...
☒	#29359	Succeeded	Interest	07:04 AM	Feb 14, 2023	 Emily Johnson	\$9	...
☐	#29359	Pending	Interest	09:50 PM	Feb 14, 2023	 Ryan Brown	\$4	...
☐	#29359	Succeeded	Payment for invoice	12:09 AM	Feb 14, 2023	 Brian Hall	\$49	...

SIGMA PAY

Dashboard

Budget Plan

Payments

Card

Saving Groups

Goals

Investments

Learn

Report and Analysis

Offers and partnerships

Settings

Payments

All payments\$1,380234 records

Succeeded\$2,380234 records

2 selected payments | \$82 total amount

	CODE	STATUS	DESCRIPTION
☐	#29359	Incomplete	Payment
☒	#29359	Succeeded	Payment
☒	#29359	Succeeded	Interest
☐	#29359	Pending	Interest
☐	#29359	Succeeded	Payment
☐	#29359	Failed	Interest
☐	#29359	Pending	Payment
☐	#29359	Succeeded	Service
☐	#29359	Failed	Payment
☐	#29359	Failed	Payment

Show 1 to 10 of 234 results

New payment

Payment Information

One time

Recurring

Your payment will be processed today May 27,2023

Change

Amount

USD | 0,0

Customer

Select or add customer

Description

Type description

Payment method

VISA

3742-4545-5400-126

David Smith

MM/YY CCV

Set default card for this customer

More option

Sent a payment email notification to the customer

Sent a SMS to the customer

Fee: \$0.5

Save

Submit

 Dashboard

 Budget Plan

 Payments

 Card

 Saving Groups

 Goals

 Investments

 Learn

 Report and Analysis

 Offers and partnerships

 Settings

Saving Groups



Monthly payment ▾

Turn ▾

Months ▾

🔍

Make a Group

View My Groups

Group 1

Enroll

Group 2

Enroll

Group 3

Enroll

Group 4

Enroll

Group 5

Enroll

Group 6

Enroll

Dashboard

Budget Plan

Payments

Card

Saving Groups

Goals

Investments

Learn

Report and Analysis

Offers and partnerships

Settings

Goal setter

📄 ⬇️ Export

Individually

Shared

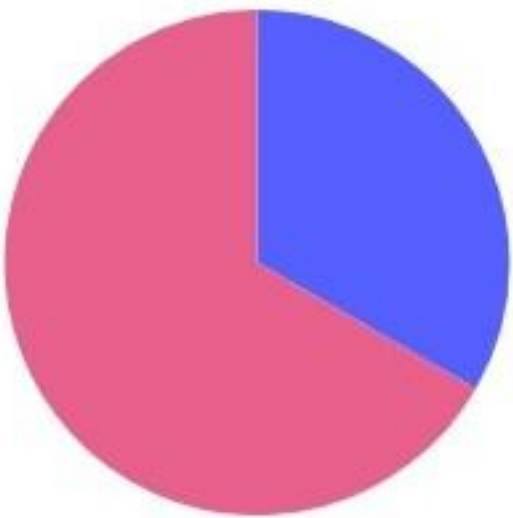


Create a goal

Achieve it with Sigma Pay

Your Free Margin

Label 1 Label 2



Sigma Pay's recommended plans



EGP 705,886
Secure my children's
education

12 Years 1,500/month



EGP 2,400,000
Save for marriage

40 Years 5000/month



EGP 7,000,000
Save for Porsche 911

40 Years 5000/month

Saving Groups

📄 ⌵ Export

Group 1 Group 2 GGroup 3

Group Details



Withdraw Date

Next Payment Date

People



PRODUCT/SERVICE	PERSON / COMPANY	EMAIL	DATE		TOTAL	
1-Month Subscription	 Jennifer Robinson	jennifer-robinson@yahoo.com	May 02, 2023	New	\$30	⋮
1-Month Subscription	 Matthew Moore	matthew-moore@hotmail.com	May 02, 2023		\$30	⋮
3-Month Subscription	 Grace Young	grace-young@hotmail.com	May 01, 2023	New	\$55	⋮
3-Month Subscription	 Emma Johnson	emma-johnson@yahoo.com	May 01, 2023	New	\$55	⋮
1-Year Subscription	 Joshua Davis	joshua-davis@hotmail.com	May 01, 2023	New	\$300	⋮

 Dashboard

 Budget Plan

 Payments

 Card

 Saving Groups

 **Goals**

 Investments

 Learn

 Report and Analysis

 Offers and partnerships

 Settings

Goals



↓ Export

Goal 1 Goal 2 Goal 3

Goals balance

EGP 1000,000.00

Remaining Amount

EGP 997,917.00

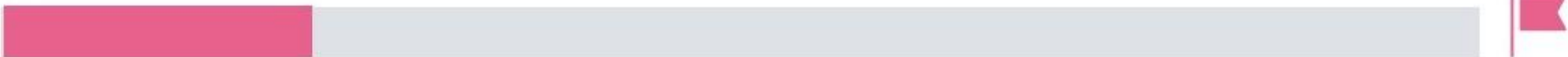
Raised amount

EGP 2,083.00

Next Due

30 Days

+ Pay



Pay in Advance

+ New payment

 Dashboard

 Budget Plan

 Payments

 Card

 Saving Groups

 Goals

 Investments

 **Learn**

 Report and Analysis

 Offers and partnerships

 Settings

Learn



Recommended for you

[View all >](#)



Video Beginner Level

Mastering Budgeting and Saving

★ 4.5 (1253) **\$39**



Photography

Smart Money Management

★ 4.5 (1233) **\$59**



Marketing 20% Off

Investing for Beginners

★ 4.5 (123) **\$39**

Personal Finance Fundamentals

Qui aliquip quis magna non sint voluptate officia qui.
Laborum sit mollit id sint et dolore conseq

Explore more



Today's Quote.

"If you wish to get rich, save what you get. A fool can earn money; but it takes a wise man to save and dispose of it to his own advantage."



Popular courses

[View all >](#)



Graphic Design Best-seller

Financial Freedom Blueprint

★ 4.5 (1253) **\$39**



Graphic Design Best-seller

Debt-Free Living Strategies

★ 4.5 (1233) **\$59**



Graphic Design Best-seller

Money Mindset Mastery

★ 4.5 (123) **\$19**

Dashboard

Budget Plan

Payments

Card

Saving Groups

Goals

Investments

Learn

Report and Analysis

Offers and partnerships

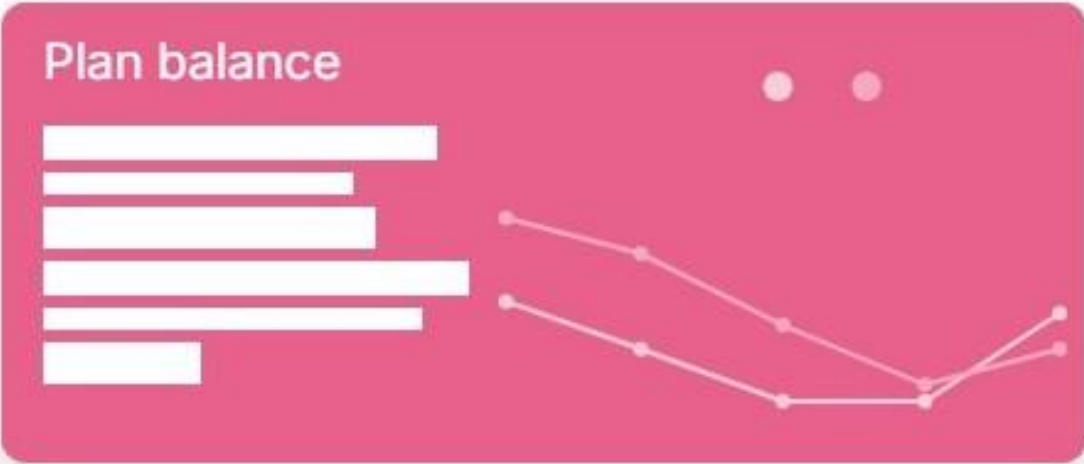
Settings

Report and Analysis

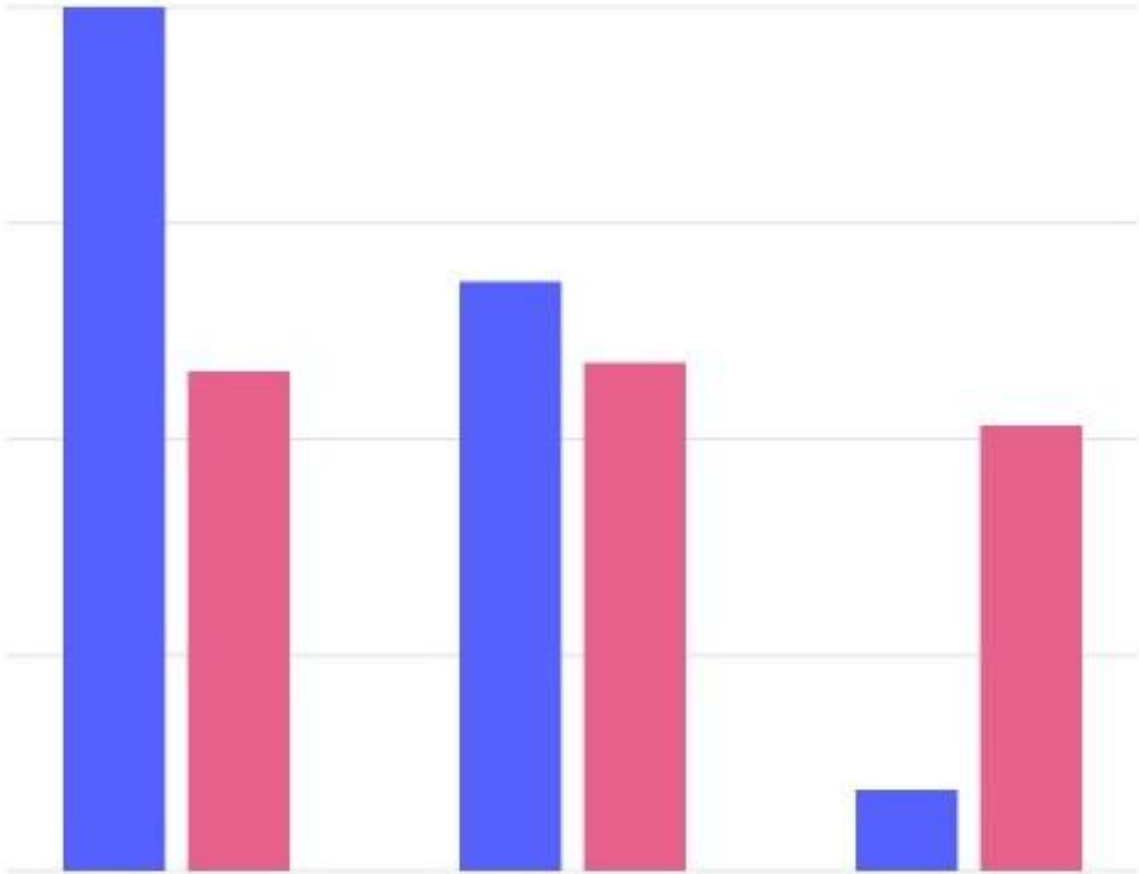


Export

Financial Report



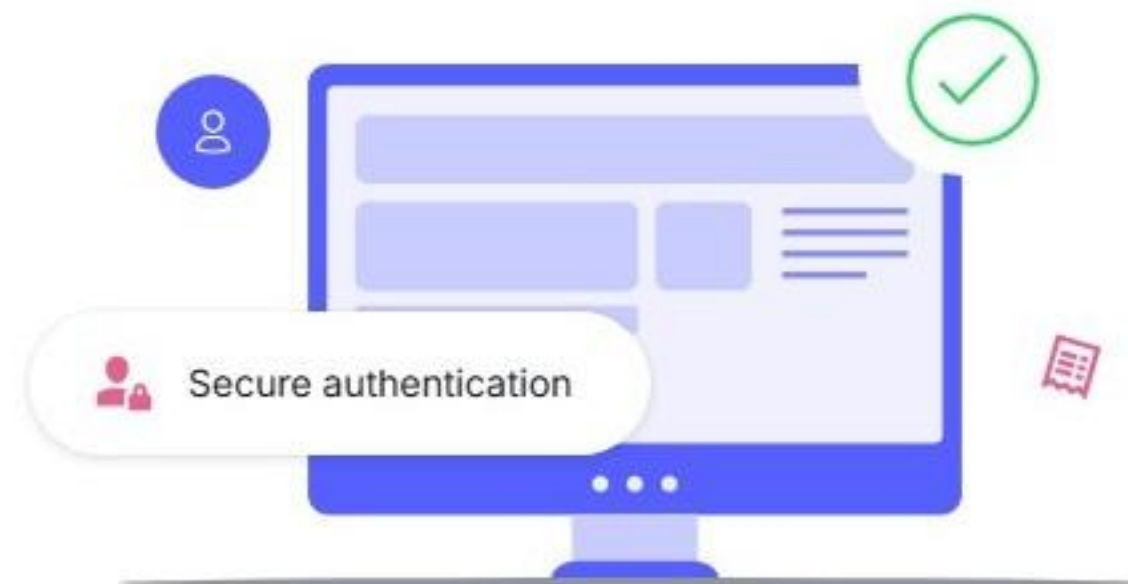
Financial Insights



☒ PDF ☒ EXCEL

Export ▾

☐		NAME	STATUS	PERIOD	DATE	UPDATED	
☐	1-Month Subscription	1-Month Subscription	Active	Monthly	Feb 15, 2022	Feb 15, 2023	...
☐	3-Month Subscription	3-Month Subscription	Active	Quarterly	Feb 15, 2022	Feb 15, 2023	...
☐	1-Year Subscription	1-Year Subscription	Active	Yearly	Feb 15, 2022	Feb 15, 2023	...



Activated account

Your account has been successfully activated.
Get ready to explore and enjoy our services.

[Go to dashboard \(20s\)](#)



We promise to prioritize the privacy and security of your personal information. We implement industry-standard security measures to protect your data from unauthorized access, loss, or misuse.

 Dashboard

 Budget Plan

 Payments

 Card

 Saving Groups

 **Goals**

 Investments

 Learn

 Report and Analysis

 Offers and partnerships

 Settings

Offers and partnerships

100%

Partnerships



Buy now
get 25% disscount

Visit



Buy now
get 5% disscount

Visit



Buy now
get 17% disscount

Visit



Buy now
get 10% disscount

Visit

View More >

Referral

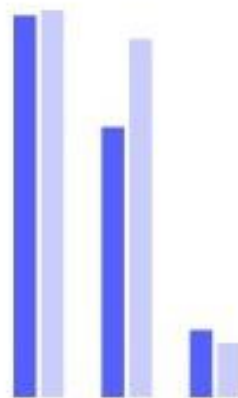
Share Link ▾

PERSON /	EMAIL	DATE	
 Jennifer Robinson	jennifer@yahoo.com	May 02, 2023	⋮
 Matthew Moore	matthew@hotmail.com	May 02, 2023	⋮
 Grace Young	grace@hotmail.com	May 01, 2023	⋮

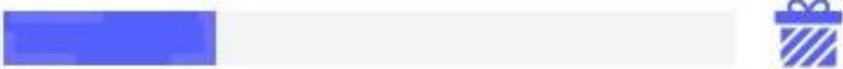
Invited Last Month

230

▲ 5.39%



Next Prize



Dashboard

Budget Plan

Payments

Card

Saving Groups

Goals

Investments

Learn

Report and Analysis

Offers and partnerships

Settings

Card

Pay with Our Card



Order now



View Card Details

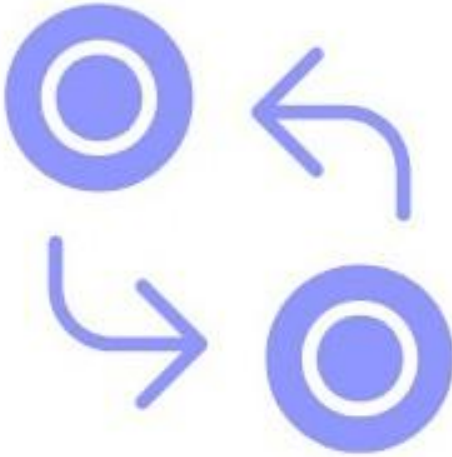
		NAME	STATUS	CHARGE	PERIOD	CREATE DATE	UPDATED	
☐	1 Month Subscription	1-Month Subscription	Active	\$30	Monthly	Feb 15, 2022	Feb 15, 2023	...
☐	3 Month Subscription	3-Month Subscription	Active	\$55	Quarterly	Feb 15, 2022	Feb 15, 2023	...
☐	1 Year Subscription	1-Year Subscription	Active	\$300	Yearly	Feb 15, 2022	Feb 15, 2023	...

- Dashboard
- Budget Plan
- Payments
- Card
- Saving Groups
- Goals
- Investments**
- Learn
- Report and Analysis
- Offers and partnerships
- Settings

INVESTMENTS

FOR ME ALL

Margin\$6,000



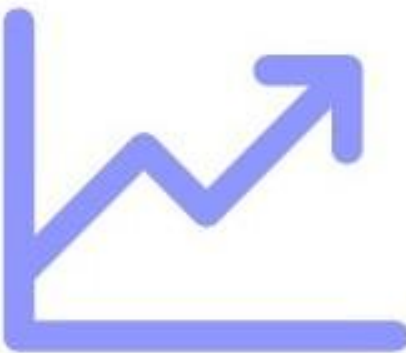
currency

Know More



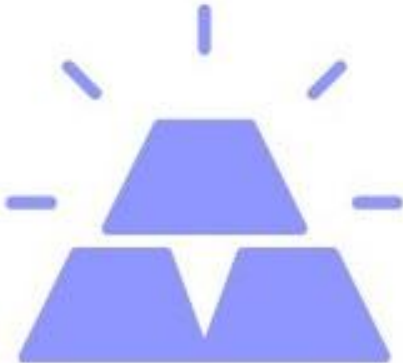
oil

Know More



stocks

Know More



Gold

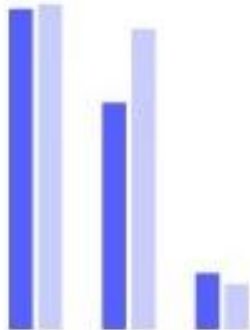
Know More

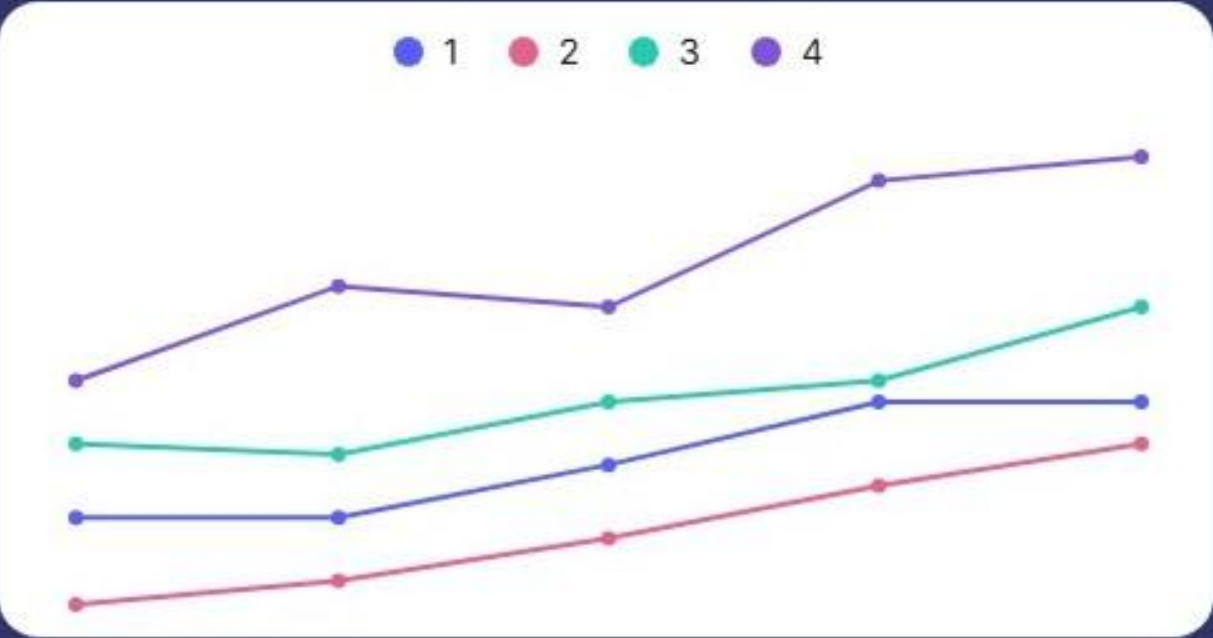
My Investments

Total Month Profit

\$730

▲ 5.39%





Cash Inflows

🕒 5mins ago

\$830

Cash Outflows

🕒 5mins ago

\$-100

Holding1	+200
Holding2	+300
Holding3	+300
Holding4	+300